

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

MA2008-1 – PROBABILITY THEORY AND NUMERICAL METHODS

(For EE, EC, AC ,VL)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Statistical tables may be provided.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. If $(x, p(x))$ values are $(0, k), (1, 2k), (2, 3k), (3, 4k)$, then value of k is
A) 0.5
B) 2
C) 0.1
D) 10
2. If $V(X) = 2$ then $V(3X + 2)$ is
A) 54
B) 18
C) 12
D) 6
3. If X and Y are independent random variables then $COV(X, Y)$ is
A) $E(XY)$
B) $E(X)E(Y)$
C) 1
D) 0
4. If $(x, p(x))$ values are $(0, 0.5), (1, 0.2), (2, 0.1), (3, 0.2)$ then $P(0 < X < 2)$ is
A) 0.2
B) 0.3
C) 0.7
D) 0.8
5. In Binomial distribution if $n=6$ and $p=0.1$ then mean is
A) 0.06
B) 0.7348
C) 0.54
D) 0.6
6. If λ is the mean of Poisson distribution then $P(X = 0)$ is given by
A) $e^{-\lambda}$
B) e^{λ}
C) e
D) 0
7. The mean of an exponential distribution is 2 then standard deviation is
A) 0.5
B) 2
C) 4
D) 0.25
8. For standardized normal distribution standard deviation is
A) 0
B) 0.5
C) 1
D) 0.25
9. While using Newton forward difference interpolation formula, for any x , the value of p is
A) $\frac{x - x_n}{h}$
B) $\frac{x - x_0}{h}$
C) $\frac{x + x_0}{h}$
D) $\frac{x + x_n}{h}$
10. Expansion of $\Delta^3 y_0$ is
A) $y_3 - 3y_2 + 3y_1 - y_0$
B) $y_3 + 3y_2 + 3y_1 + y_0$
C) $y_4 + 3y_3 + 3y_2 + y_1$
D) $y_4 - 3y_3 + 3y_2 - y_1$

11. First order divided difference of tabular points (5, 150) and (7, 392) is
 A) 242 B) 2
 C) 221 D) 121
12. If $f(x) = x^3 + 5x - 7$, then $\Delta^4 f(x)$ is
 A) 6 B) 1
 C) 0 D) 12
13. If $x_0, x_1, x_2, \dots, x_n$ are equally spaced, then expression for $\frac{dy}{dx}$ at $x = x_n$, using backward difference operator is
 A) $\frac{1}{h} \{\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots\}$ B) $\frac{1}{h^2} \{\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n \dots\}$
 C) $\frac{1}{h} \{\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n \dots\}$ D) $\frac{1}{h^2} \{\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots\}$

14. For the data:

x	2	4	6	8
$f(x)$	3	5	6	7

$\int_2^8 f(x) dx$ when found by the Trapezoidal rule is

- A) 16 B) 30
 C) 45 D) 32
15. Given $(n+1)$ set of tabular values (t, v) of time and velocity, then we can obtain the acceleration of the particle by process of
 A) Interpolation B) Extrapolation
 C) Numerical differentiation D) Numerical integration
16. To apply Simpson's $\frac{1}{3}$ rd rule, the interval (x_0, x_n) must be divided into _____ number of sub intervals.
 A) odd B) even
 C) multiples of 3 D) any
17. In Runge-Kutta method of order four, the formula to find k_4 is
 A) $k_4 = hf(x_0 + h, y_0 + k_3)$ B) $k_4 = hf(x_0 + \frac{h}{2}, y_0 + k_3)$
 C) $k_4 = hf(x_0 + h, y_0 + \frac{k_3}{2})$ D) $k_4 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_3}{2})$
18. The root of the equation $xe^x - 3 = 0$ lies in the interval
 A) (0,1) B) (-1,0)
 C) (2,3) D) (1,2)
19. For the heat equation $u_t = \frac{1}{2} u_{xx}$, $h = k = 1$, the value of mesh ratio parameter α is
 A) 1 B) 0.5
 C) 0.25 D) 2
20. The standard five-point formula for solving Laplace's equation $u_{xx} + u_{yy} = 0$ is
 A) $u_{i,j} = u_0 + \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$ B) $u_{i,j} = \frac{1}{3} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$
 C) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$ D) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}] - u_{0,0}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I****Marks BT* CO* PO***

1. a) The probability distribution of a finite random variable X is given by the following table:

X	-2	-1	0	1	2	3
$P(X)$	0.1	k	0.2	$2k$	0.3	k

Find (i) The value of k (ii) $P(X > -1)$ (iii) $E(X)$.

6 L*1 1 2

- b) State and prove Bayes' theorem.

6 L2 1 1

- c) 6 coins are tossed, find the probability of getting (i) exactly 3H (ii) at most 3H.

4 L1 2 2

2. a) Let X be a continuous random variable with p.d.f. given by

$$f(x) = \begin{cases} x & , 0 \leq x < 1 \\ 2 - x & , 1 \leq x < 2 \\ 0 & , \text{elsewhere} \end{cases}$$

Find the c.d.f. $F(x)$.

6 L1 1 2

- b) Obtain the mean and variance of an Exponential distribution.

6 L2 2 1

- c) If the probability that an individual suffers a bad reaction from an injection is 0.001. Using Poisson distribution, determine the probability that out of 2000 individuals (i) exactly 3 (ii) more than 1 individual suffers a bad reaction.

4 L1 2 2

3. a) A joint distribution of (X, Y) is given by the following table.

$Y \backslash X$	-4	2	7
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
5	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

Determine (i) marginal distributions of X & Y . (ii) Test the independence of X & Y . (iii) $COV(X, Y)$.

6 L1 1 2

- b) Suppose that X has distribution $N(2, 0.16)$. Evaluate (i) $P(X \geq 2.3)$ (ii) $P(1.8 \leq X \leq 2.1)$.

6 L2 2 2

- c) Mr. Sharma who lives in the outskirts of a city wants to catch an early morning train. The probability that he gets an auto at that time is only 0.23. If he gets an auto, the probability of catching the train is 0.85. If he fails to get an auto, he has to take a city bus and probability he gets train is only 0.43. Find the probability that he catches the train.

4 L1 1 2

Unit – II

4. a) Using Newton's forward difference interpolation formula, compute the pressure of the steam at the temperature of 142° from the steam table.

Temperature	140	150	160	170	180
Pressure	3.685	4.854	6.302	8.076	10.225

6 L1 3 2

- b) Using Newton's divided difference formula, find $f(8)$ from the following data.

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

6 L1 3 2

- c) Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ by using Simpson's $\frac{1}{3}$ rd rule taking four equal strips.
5. a) A function $y = f(x)$ is given by the following table.
- | | | | | | | |
|-----|-----|-------|-------|-------|-------|------|
| x | 1.0 | 1.2 | 1.4 | 1.6 | 1.8 | 2.0 |
| y | 0 | 0.128 | 0.544 | 1.296 | 2.432 | 4.00 |
- Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at 1.0 by suitable interpolation formula.
- b) Derive the general quadrature formula and hence deduce Trapezoidal rule.
- c) Find the value of y at $x = 2$ using Lagrange's interpolation formula from the following data:

x	0	1	3
y	-12	0	6

6. a) From the given table, estimate the number of students who obtained marks between 40 and 45.

Marks	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
Number of students	31	42	51	35	31

- b) The velocity $v(\text{km/min})$ of a moped which starts from rest is as follows:

t	2	4	6	8	10	12	14	16	18	20
v	10	18	25	29	32	20	11	5	2	0

Estimate the distance covered in 20 minutes.

- c) Find the missing term in the following table using finite differences.

x	0	1	2	3	4
y	1	3	9	---	81

Unit – III

7. a) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown below:

	0	0	
0			2
0			2
	2	2	

- b) Using modified Euler's method, find an approximate value of y when $x = 0.2$ given that $\frac{dy}{dx} = x + y$ and $y(0) = 1$. Take $h = 0.1$ and carry out two iterations in each step.
- c) Using Regula-Falsi method find an approximate root of the equation $x^3 - 2x - 5 = 0$ in the interval $(2,3)$. Carry out two iterations.
8. a) Using fourth order Runge-Kutta method solve the initial value problem $\frac{dy}{dx} = x + y^2, y(0) = 1$, at $x = 0.1$ taking $h = 0.1$.
- b) Solve the PDE $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square with sides $x = 0 = y; x = 3 = y$ with $u = 0$ on the boundary and $h = 1$.
- c) Find the solution of the equation $u_{xx} = 2u_t$ with $u(0, t) = u(4, t) = 0$ and $u(x, 0) = x(4 - x)$, taking $h = 1, k = 1$. Find the values up to $t = 3$.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
